

Introduction to AI Agents

What is an LLM?

A Large Language Model (LLM) is a type of artificial intelligence trained on massive amounts of text data. LLMs can understand and generate human language. Popular examples include GPT-4, Claude, and LLaMA. They are used in chatbots, code generation, summarization, and question answering.

What is RAG?

RAG stands for Retrieval-Augmented Generation. It is a technique that combines a retrieval system with a language model. Instead of relying only on the model's training data, RAG first searches a knowledge base for relevant documents, then passes those documents along with the user query to the LLM to generate a grounded answer. This reduces hallucinations and allows the model to answer questions about private or recent documents.

What are Embeddings?

Embeddings are numerical representations of text. Each sentence or paragraph is converted into a vector of numbers. Sentences with similar meanings have vectors that are close together in vector space. Cosine similarity is used to measure how similar two vectors are. Embeddings are the core technology behind semantic search and RAG retrieval.

What is a Vector Database?

A vector database stores embeddings and allows fast similarity search. When a user query arrives, it is converted into an embedding and compared against all stored document embeddings. The most similar documents are returned as context. Popular vector databases include ChromaDB, FAISS, Pinecone, and Weaviate. ChromaDB is free, local, and easy to use for beginners.

What is Tool Calling?

Tool calling allows an LLM to invoke external functions or APIs during a conversation. The model decides when a tool is needed, formats the correct arguments, and uses the result in its response. Common tools include calculators, weather APIs, search engines, and database queries. Tool calling is the foundation of autonomous AI agents.

What is an AI Agent?

An AI agent is a system where an LLM autonomously decides what actions to take to complete a goal. It uses a reasoning loop: observe the situation, decide on an action, call a tool, observe the result, and repeat. This is called the ReAct pattern (Reason and Act). Agents can browse the web, write and execute code, send emails, and manage files.